Client Briefing Report

BUSA8031 Assignment 2

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Introduction

This report provides a data-driven analysis of population trends in Australia from 1996 to 2016, utilising data from the Australian Bureau of Statistics (ABS) and presented through interactive dashboards developed in Power BI. This analysis aims to facilitate clients in swiftly recognising, comprehending, and responding to significant demographic shifts influencing Australian society. The report has three dashboards, each designed for a distinct analytical objective. Collectively, they offer a thorough and nuanced perspective on population dynamics, encompassing overarching national trends as well as intricate insights into specific countries of birth (COBs) and the underlying narratives within the data.

Dashboard 1 offers a comprehensive national population overview, detailing temporal changes in overall population, state/regional growth, gender distribution, and age group demographics. This dashboard aids in contextualising the overarching framework and identifying macro-level trends that may influence infrastructure, services, and workforce planning.

Dashboard 2 concentrates on examining population trends based on country of birth. This analysis examines demographic changes across gender, temporal factors, and regional distribution across a chosen cohort of ten nations, including Australia. This dashboard facilitates more sophisticated analysis and emphasises demographic diversity and disparities among migrant populations.

Dashboard 3 employs a narrative technique by focusing on a particular revelation revealed through comprehensive data analysis: the remarkable percentage increase in distinct smaller migrant populations from 2001 to 2016. Specifically, it indicates that nations such as Nepal, Bhutan, and Liberia witnessed significant surges in their migratory populations, although commencing from a minimal baseline. This research reveals subtle yet significant trends that may otherwise be undetected.

These dashboards were created not merely to display data, but to enhance decision-making by integrating statistical rigour with effective visual communication. Power BI's interactivity enabled end-users to filter data by area, time, gender, and country of birth, ensuring that the dashboards are customisable for various business or policy contexts.

Tools and Methodology

All visualisations in this report were developed using Power BI Web, which offers strong data visualisation and interactive dashboarding capabilities. As the analysis was conducted on a Mac device, the use of Power BI Desktop was not available, leading to some limitations in advanced functionality such as calculated measures or DAX-based transformations. Nevertheless, creative workarounds using filters, slicers, and pre-aggregated fields allowed for effective storytelling and exploratory insight generation across all three dashboards.

Dashboard 1: Population Overview

The first dashboard provides a national-level overview of Australia's population trends between 1996 and 2016. It focuses on three core demographic attributes: time, age group, and gender, broken down by state or region. This dashboard was designed to answer foundational questions such as: How has Australia's total population changed over the last two decades? Are certain states growing faster than others? How do gender and age distributions shift over time?

The dashboard includes the following key visuals:

- A line chart showing total population growth across all regions over time,
- A stacked bar chart depicting population by age group and sex
- A clustered bar chart comparing regional/state-level populations.

To improve user engagement and insight discovery, slicers were added to allow dynamic filtering by:

- Region (e.g., New South Wales, Victoria, Queensland),
- Sex (Male/Female),
- Year (from 1996 to 2016).

These interactive filters enable users to explore how population structures evolve over time, with a focus on the practical implications for service delivery, infrastructure planning, and resource allocation.

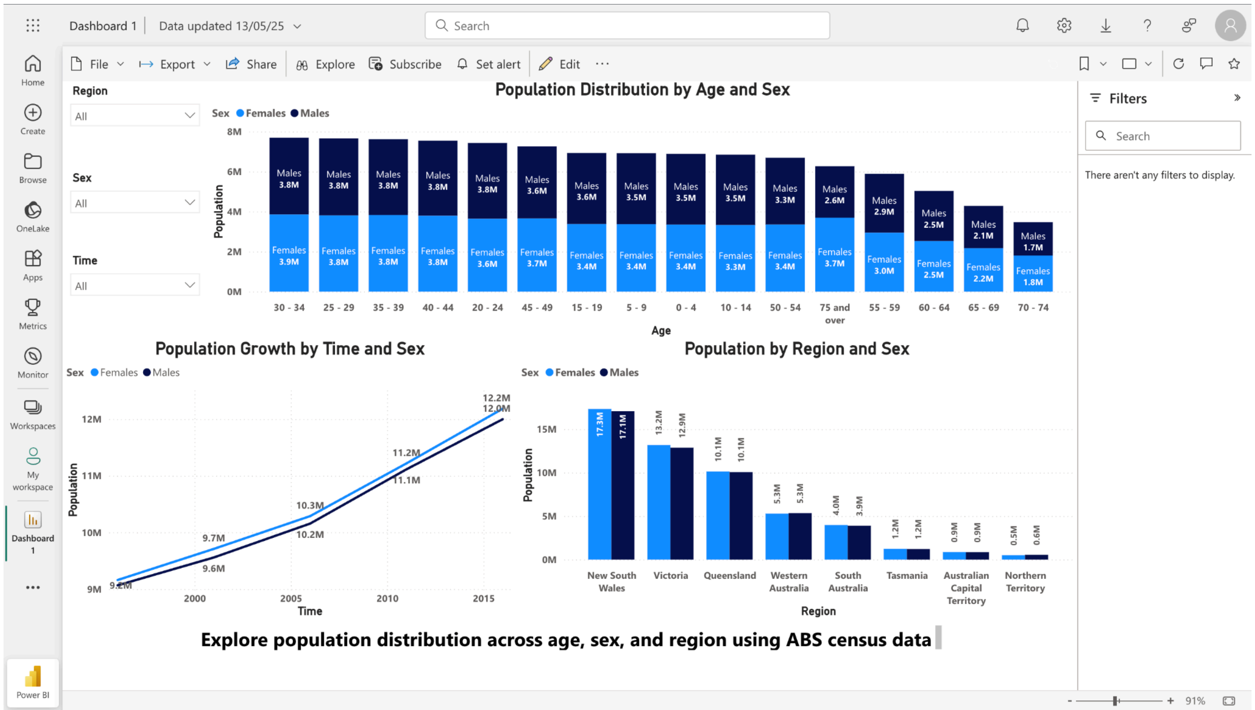
Key Observations:

- National population increased steadily from 1996 to 2016, with particularly strong growth in New South Wales and Victoria.
- The gender distribution remained relatively stable across the years, with a near-even split.
- The population shows signs of ageing, with increasing representation in the 50–74 age group, reflecting broader societal trends around longevity and shifting workforce needs.

This dashboard serves as a foundation for the next stages of analysis by providing the broader demographic context in which more specific insights — such as country-of-birth growth — can be interpreted.

Figure 1: National Population Overview Dashboard in Power BI

This dashboard visualises Australian population trends from 1996 to 2016 by state, age group, and gender, with interactive filters for regional and demographic exploration.



Dashboard 2: Population by Country of Birth

The second dashboard shifts focus from Australia's overall population to explore the country-of-birth composition of its residents, revealing how global migration trends have shaped the nation's demographic diversity. This dashboard answers key questions such as: Which countries contribute the most to Australia's overseas-born population? How does this distribution vary by gender and region? And what share do smaller migrant communities represent when dominant countries are excluded?

Key Visuals:

- Clustered Column Chart (Top 10 Countries of Birth): Displays total population for each country of birth, segmented by sex. This helps highlight both the volume and gender split of each group, with Australia excluded to focus solely on migrant populations.
- Pie Chart (Top 10 Countries Excluding Australia): Shows proportional share of population by country of birth, reinforcing the size of key migrant communities relative to each other.
- Horizontal Bar Chart (Population by Country of Birth and Sex): Highlights population size and gender distribution in a more space-efficient format.

- Slicers for Region, Sex, and Age: Allow users to dynamically filter visuals, uncovering differences in migration patterns by location or demographic group.

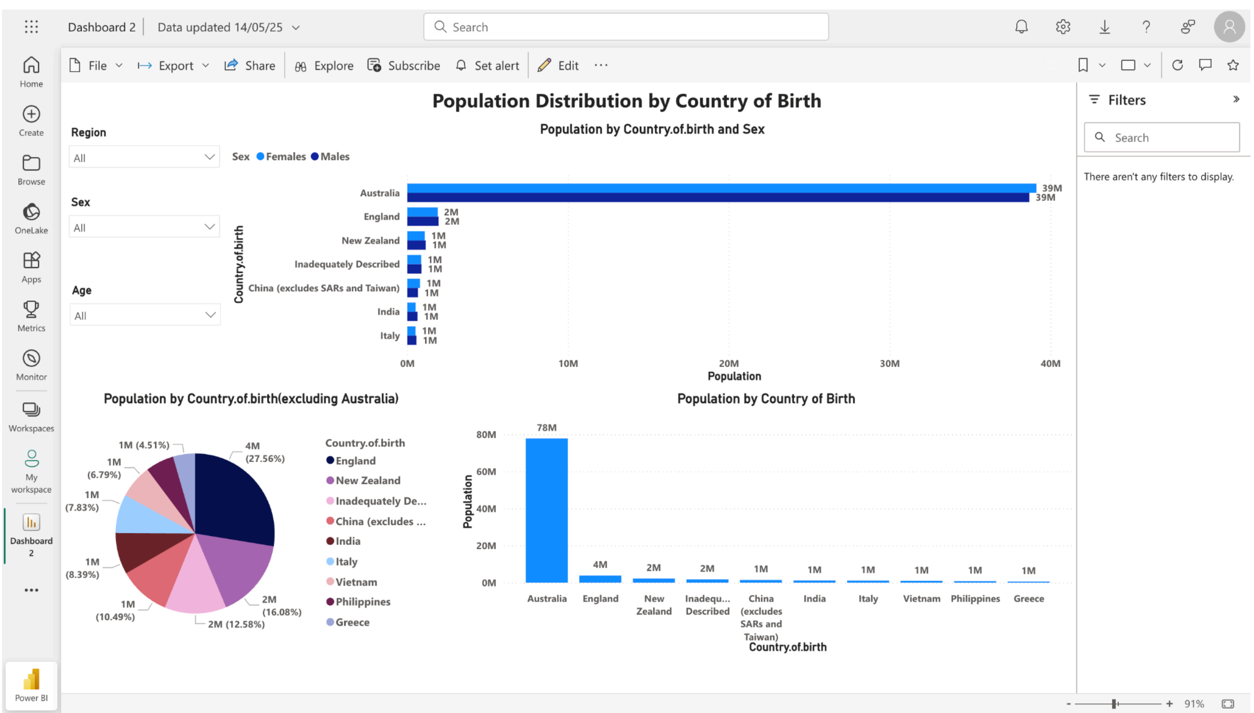
Key Observations:

- The top migrant countries include England, India, China (excluding SARs), and New Zealand, with most of these groups having relatively balanced gender splits.
- Some countries show notable gender imbalances—for example, Vietnam and Pakistan have a slightly higher proportion of males.
- Visualising by proportion (via pie chart) draws attention to the breadth of Australia's migrant base, beyond just a few dominant nations.
- Users can explore region-specific compositions using slicers, which may be particularly relevant for local policy planning.

This dashboard enhances understanding of Australia's multicultural makeup and serves as a bridge between national population patterns (Dashboard 1) and more nuanced growth analysis (Dashboard 3).

Figure 2: Country of Birth Distribution of Australian Residents (2016)

This dashboard visualises the distribution of Australia's population by country of birth, highlighting both dominant countries like Australia and England, as well as a breakdown excluding Australia to reveal migrant diversity more clearly.



Dashboard 3:Beyond the Obvious – The Fastest Growing Migrant Communities in Australia (2001–2016)

The third dashboard takes a storytelling approach to uncover a lesser-known but compelling insight: the emergence of fast-growing migrant communities in Australia, based not on size but on percentage growth over time. This perspective goes beyond the usual focus on countries with the largest migrant populations and instead highlights countries whose communities have rapidly expanded — often under the radar.

The core research question that shaped this dashboard was: Which countries of birth, despite starting from a small base, have experienced the highest relative growth in Australia between 2001 and 2016? Addressing this question required filtering out larger, dominant populations (like India, China, and the UK) and spotlighting countries with sharp growth trajectories.

Dashboard Design:

- The main line chart presents the top 10 fastest-growing migrant groups, calculated by percentage increase in population from 2001 to 2016. Countries like Bhutan, Liberia, Rwanda, Mongolia, and Kyrgyzstan showed exponential increases, some over 3000%.
- To avoid skewed scaling in the main chart, a second visual isolates Nepal, which experienced the most significant growth — from just 2,430 people in 2001 to over 59,000 in 2016 — representing an astonishing 2330% increase.
- The dashboard also includes interactive slicers for Region and Sex, allowing users to explore whether these growth trends are regionally concentrated or gender-specific.

This design enables both comparative insight across multiple countries and focused exploration of outliers like Nepal. By separating Nepal into its own chart, visual clarity is maintained, and the story remains easy to interpret.

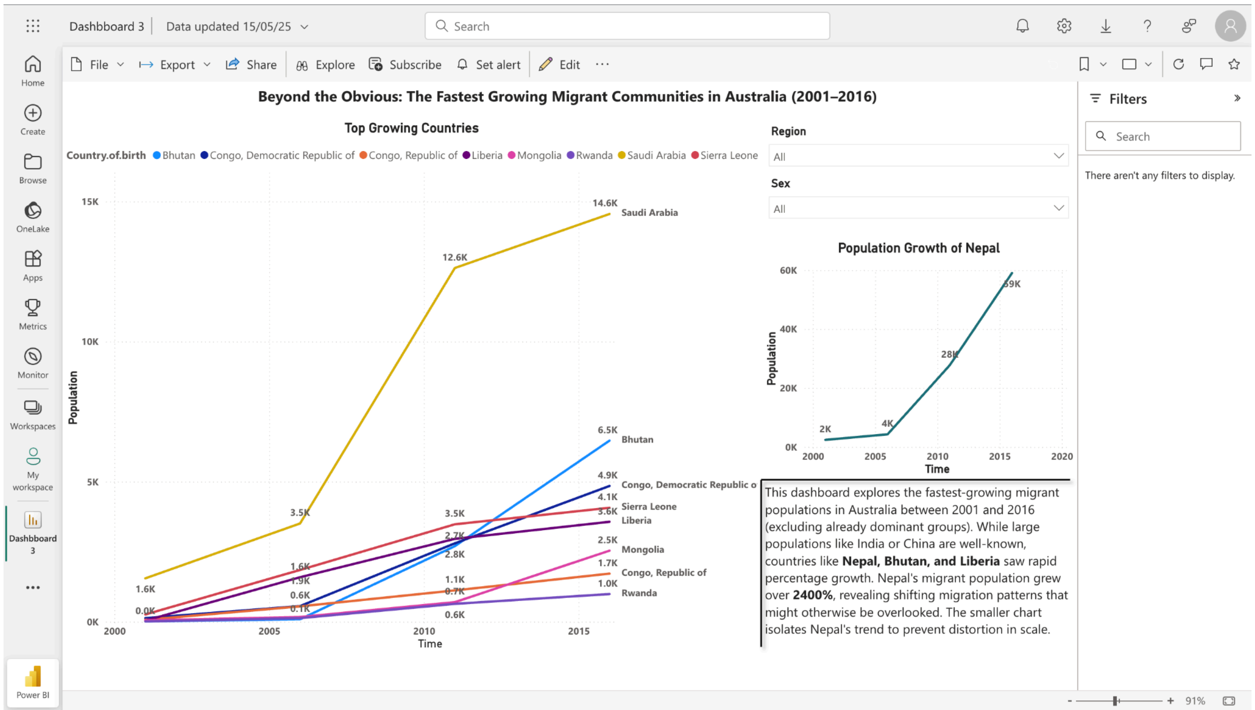
Analytical Approach:
Because Power BI Web does not support DAX or custom measure creation — a limitation of using a Mac-based environment — the percentage growth values were pre-calculated externally in Excel using standard formulas. These countries were then manually filtered and visualised in Power BI to comply with the constraints while still delivering an effective analytical narrative.

Why This Insight Matters:
This dashboard demonstrates how small, emerging communities can grow significantly over short periods, often due to humanitarian migration pathways, international education, or shifts in visa policy. These trends can influence local government decisions, multicultural support services, and long-term urban planning.

By drawing attention to patterns that may otherwise be overlooked, this dashboard fulfils the storytelling requirement by delivering not just visualisation, but a focused, surprising, and meaningful story grounded in real-world implication.

Figure 3: Top 10 Fastest-Growing Migrant Communities in Australia (2001–2016)

This dashboard identifies the highest percentage growth in country-of-birth populations from 2001 to 2016. A secondary chart isolates Nepal's trajectory, which saw over 2400% growth, for scale clarity.



Key Findings and Insights

National Demographic Trends (Dashboard 1)

- Australia's population grew steadily over the 20-year period, with New South Wales and Victoria contributing the largest absolute increases.
- Ageing is evident, with a clear rise in the 50–74 age cohort, reflecting extended life expectancy and shifting retirement demographics.
- The gender split remained stable across time and regions, confirming long-term demographic balance.

Country of Birth Composition (Dashboard 2)

- England, India, China, and New Zealand remain the largest migrant populations.
- When excluding Australia, the data reveals a diverse mix of origin countries, with relatively balanced gender representation across most migrant groups.

Emerging Growth Stories (Dashboard 3)

- Nepal was the fastest-growing migrant group in percentage terms, increasing by over 2400% from 2001 to 2016.
- Other countries like Bhutan, Liberia, and Mongolia also showed exceptional relative growth, indicating new migration patterns driven by education, humanitarian, or family reunification pathways.
- These insights underscore the importance of looking beyond top-10 totals to understand emerging communities with distinct needs and trajectories.

Collectively, these findings show how demographic insights — when segmented effectively — can inform smarter policy, targeted community programs, and a more nuanced understanding of Australia's population evolution.

Assumptions and Limitations

Several simplifications and constraints shaped this analysis:

- Data Scope: ABS census data from 1996 to 2016 was used. Post-2016 trends are not captured.
- Simplified Groupings: Age categories were grouped (e.g. 0–24, 25–49) and only the top 10 countries of birth were analysed to maintain clarity.
- Platform Constraints: Due to using Power BI Web on a Mac, advanced features like DAX measures and custom calculations were unavailable. Percentage growth for Dashboard 3 was pre-calculated in Excel and manually filtered.
- Static Rankings: Storytelling insights (e.g. fastest-growing countries) are based on fixed selections and are not dynamically updated by slicers.

These choices ensured visual clarity and focus, while allowing key insights to emerge despite technical limitations.

Conclusion

This report provides a systematic examination of Australia's population and migration trends, utilising data from the Australian Bureau of Statistics, and is illustrated through three interactive dashboards created in Power BI Web. Each dashboard was created to deliver unique yet interrelated facts, transitioning from a comprehensive national picture to localised patterns within migrant communities, and ultimately to a concentrated narrative on swift population changes.

The initial dashboard provided essential background, illustrating a consistently expanding and ageing population, with New South Wales and Victoria at the forefront of regional expansion. The second dashboard examined the varied composition of Australia's migrant community, illustrating the contributions of different countries of origin to the national fabric and highlighting the distinctions in gender and regional patterns across these groups. The resulting dashboard presented a concentrated, data-centric narrative, emphasising smaller yet swiftly growing migrant communities that could be neglected in high-level summaries.

Collectively, these dashboards demonstrate the significance of meticulously segmenting data to derive insights that transcend superficial patterns. The report illustrates how data storytelling, enhanced by interaction, may effectively convey intricate demographic information with clarity and persuasion. Despite limitations in platform capabilities and data granularity, the adopted strategy illustrates that accessible tools can yield significant insights.

The capacity to dynamically visualise change is essential for governments, community organisations, and companies. Insights like those provided here can inform decisions regarding urban planning, multicultural service delivery, and strategic communications. As Australia's demographic landscape evolves, the tools and procedures for its interpretation must also adapt.